Radxa E20C

schematic

V1.10

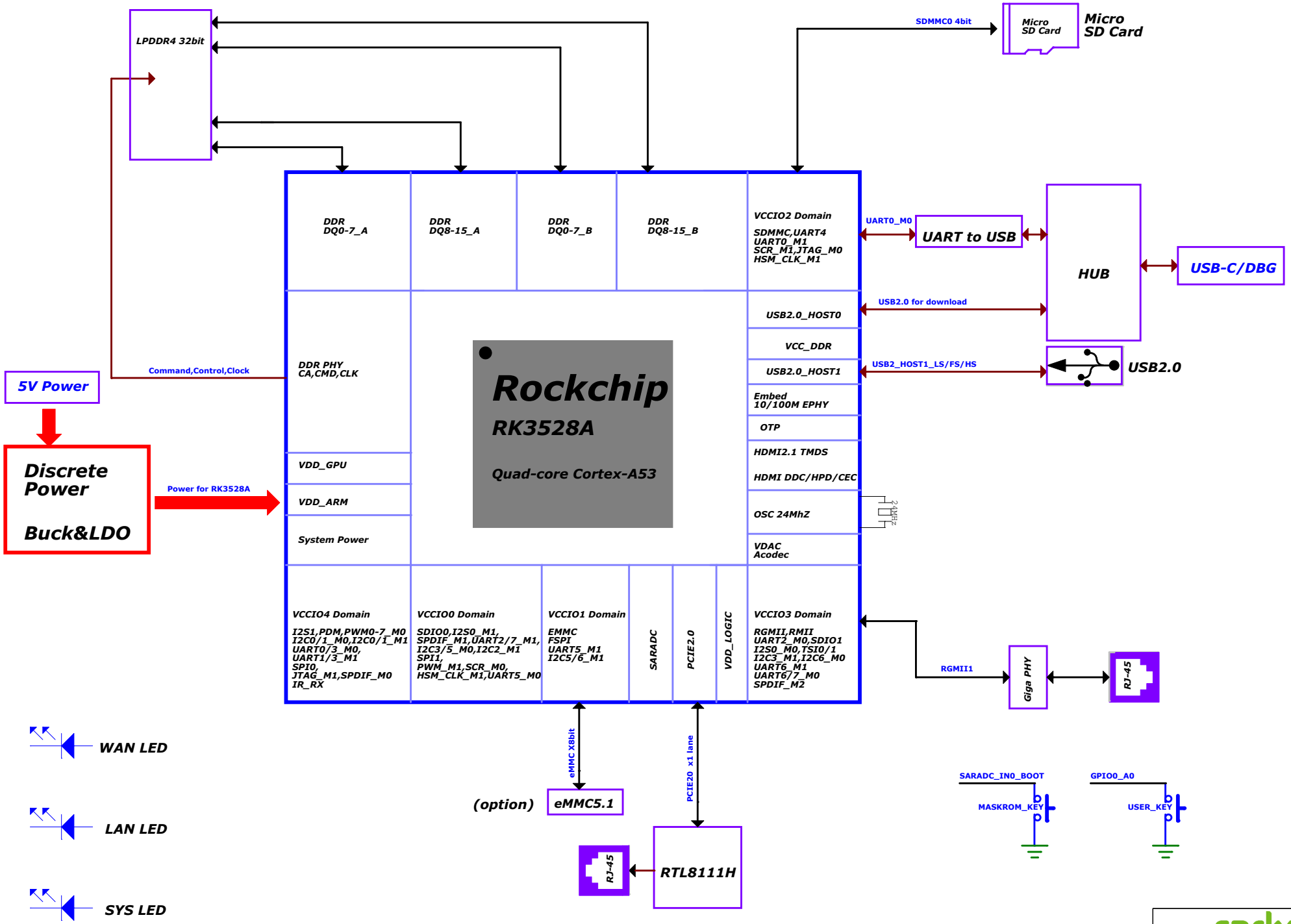
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radxa

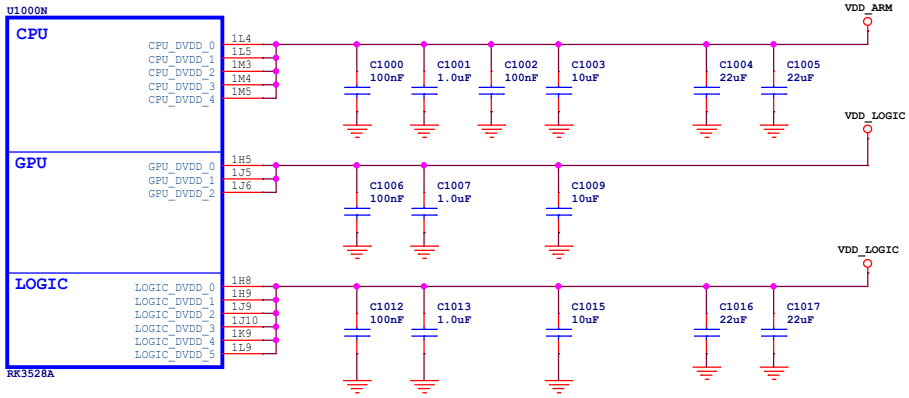
Size	Title:	REV
A4	E20C	V1.10
	Page Name: 02.Index	
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RK3528A Block Diagram

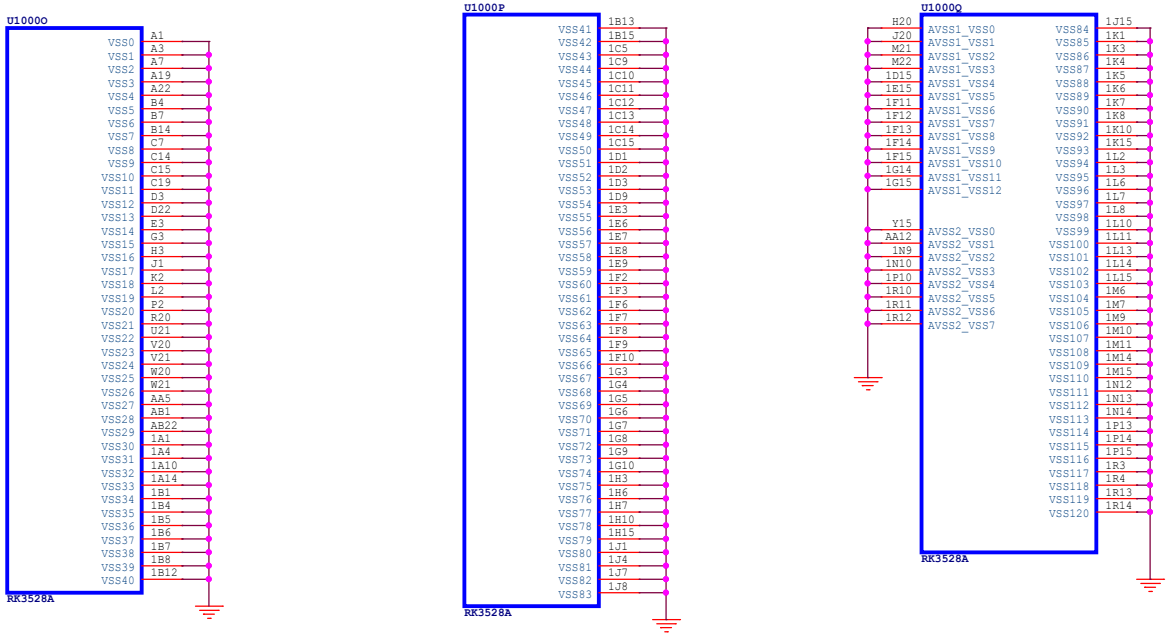


Size: A2	Title: E20C	REV: V1.10
	Page Name: 03.Block	
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RK3528A_N(POWER)

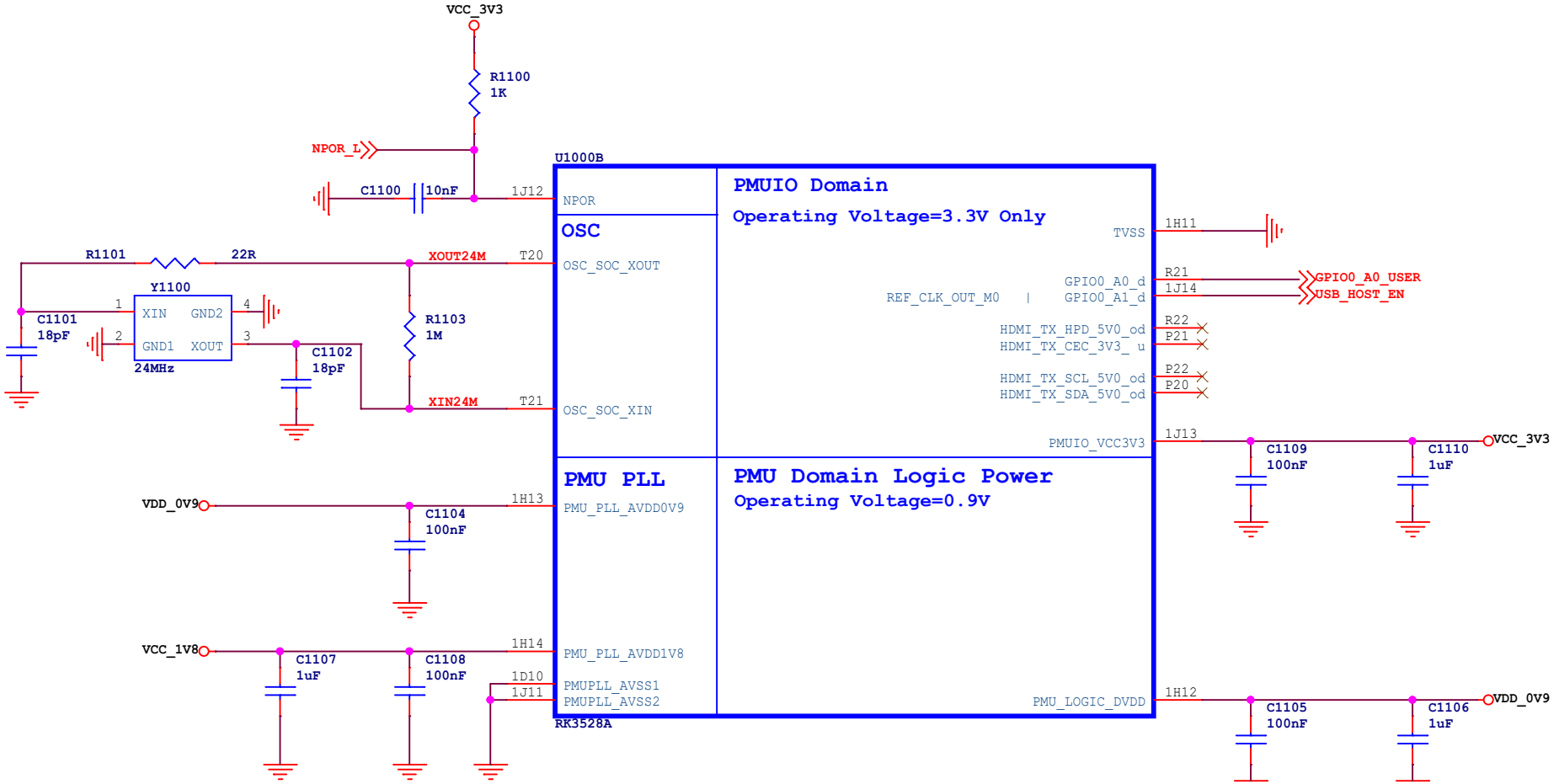


RK3528A_O/P/Q(GND)



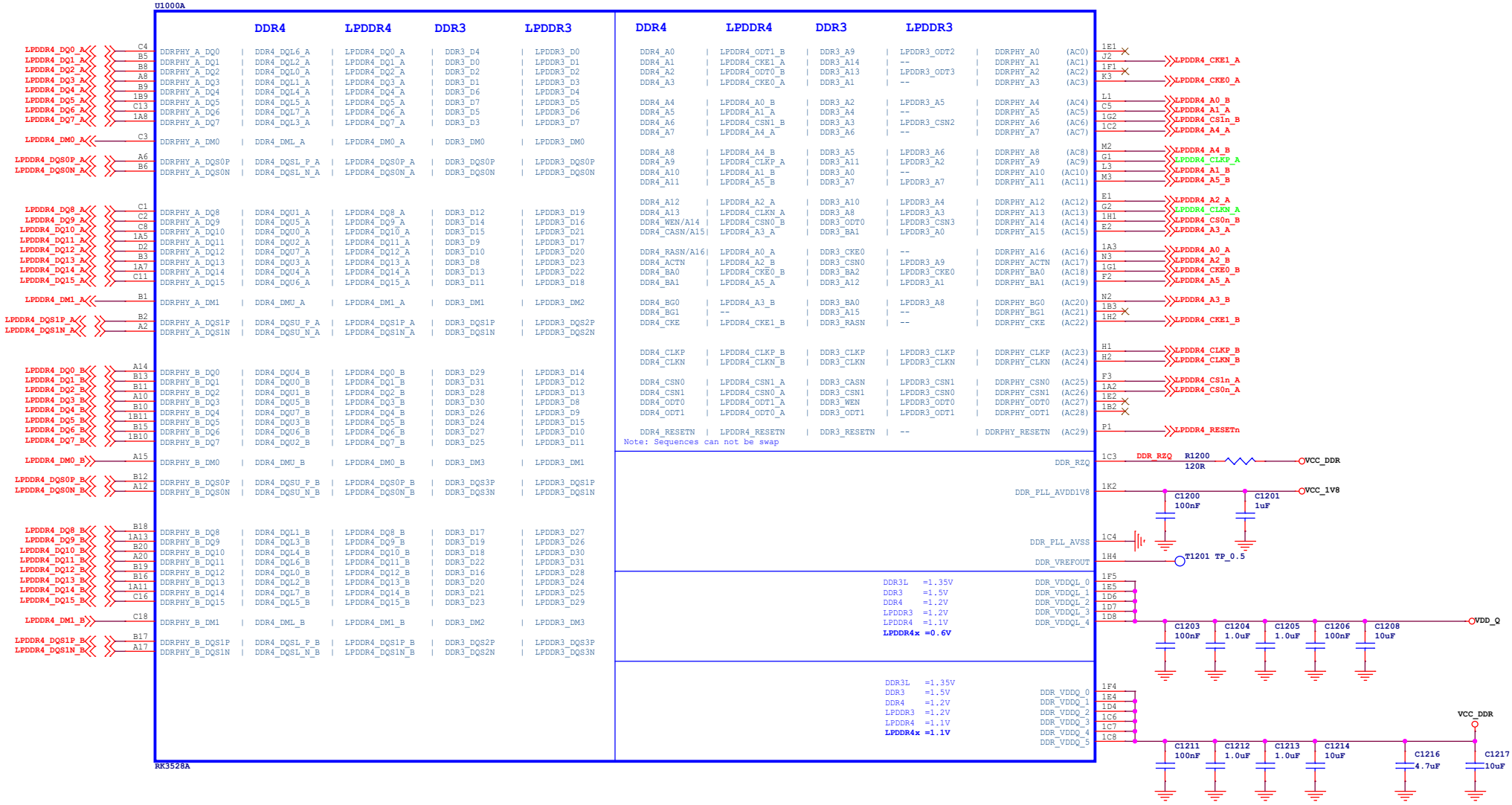
Size: A3	Title: E20C	REV: V1.10
	Page Name: 04.RK3528A Power/GND	
Date: Thursday, June 06, 2024	Sheet 04 of 22	

RK3528A_B(OSC/PLL/PMUIO Domain)



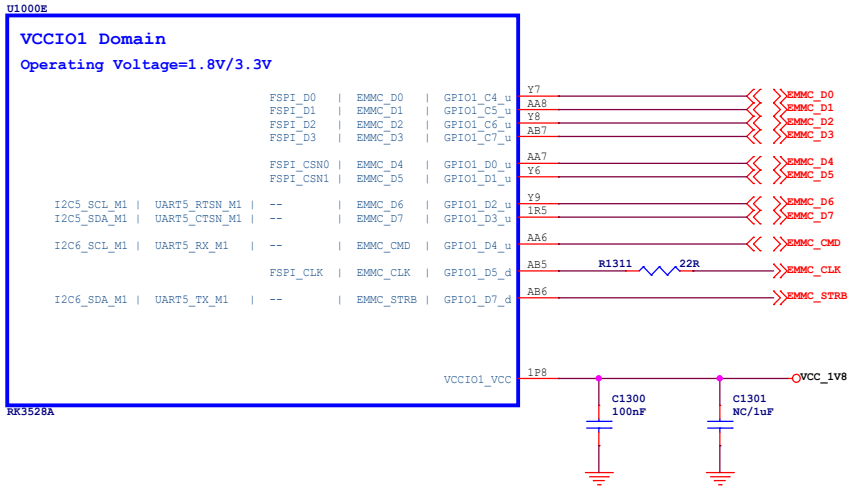
Size A4	Title: E20C	REV V1.10
	Page Name: 05.RK3528A OSC/PLL/PMUIO	
Date: Thursday, June 06, 2024	Sheet 05 of 22	

RK3528A_A(DDR PHY)

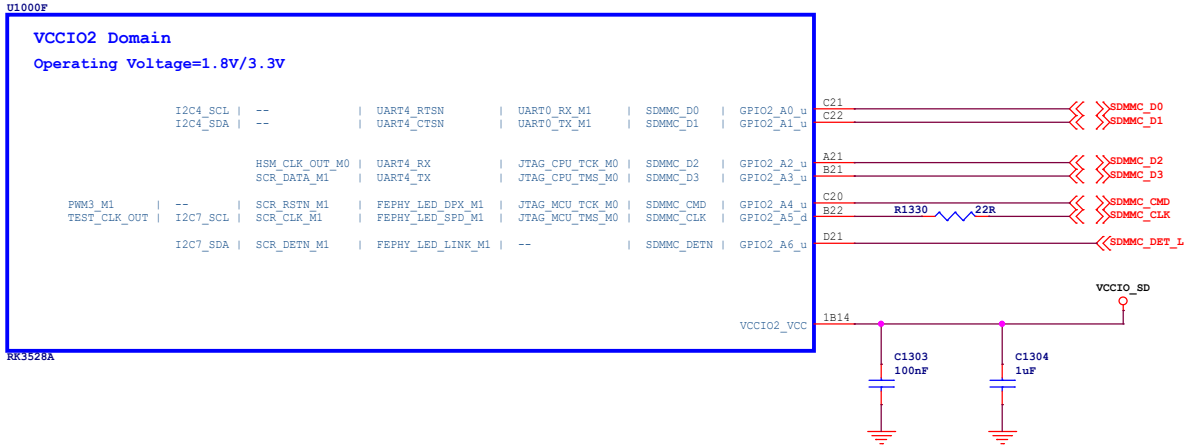


Size	Title:	REV
A3	E20C	V1.10
	Page Name: 06.RK3528A DDR Controller	
Date: Thursday, June 06, 2024	Sheet 06 of 22	

RK3528A_E(VCCIO1 Domain)

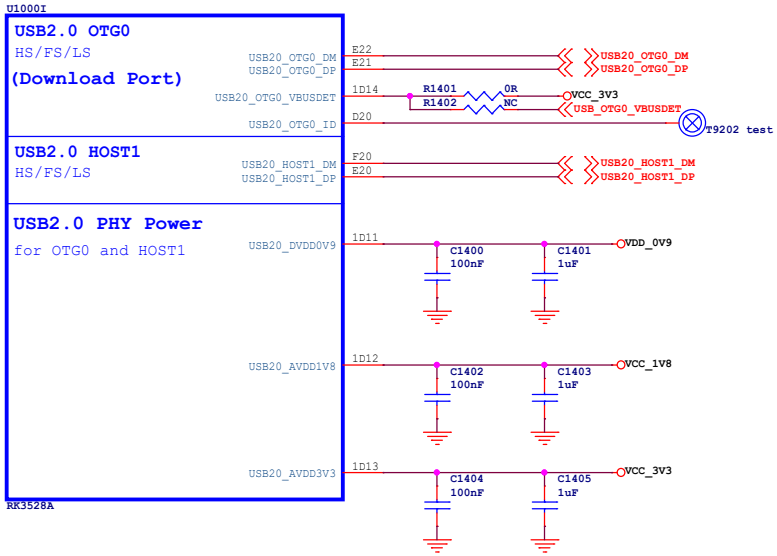


RK3528A_F(VCCIO2 Domain)

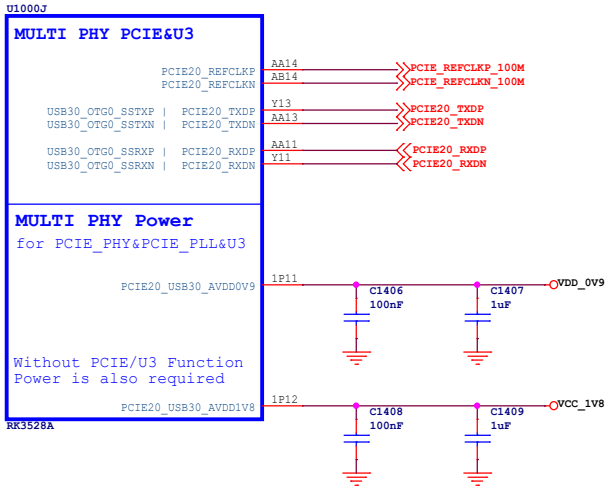


Size	Title: E20C	REV
A3	Page Name: 07.RK3528A FLASH/SD Controller	V1.10
Date: Thursday, June 06, 2024	Sheet 07 of 22	

RK3528A_I(USB2.0 OTG/HOST)

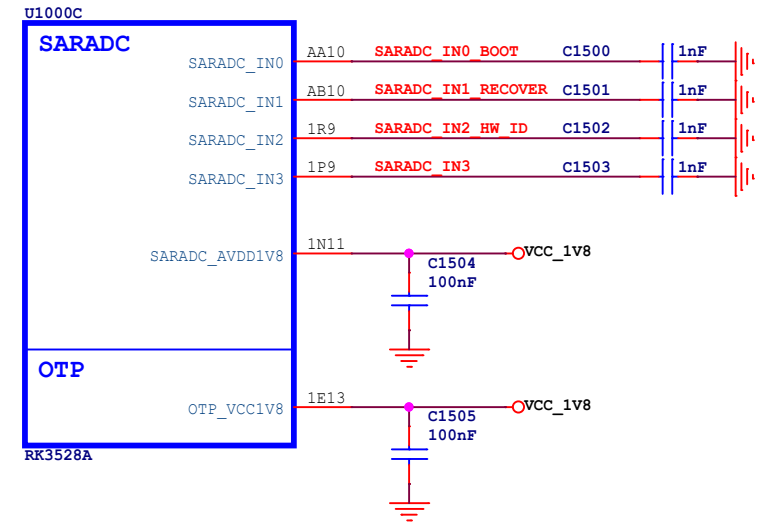


RK3528A_J(PCIE2.0/U3 PHY)



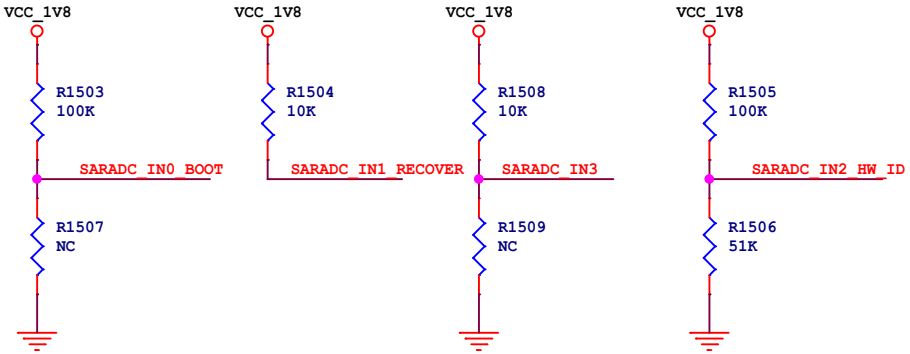
Size	Title: E20C	REV
A3	Page Name: 08.RK3528A USB/PCIE Controller	V1.10
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RK3528A_C(Saradc/OTP)



TABLE

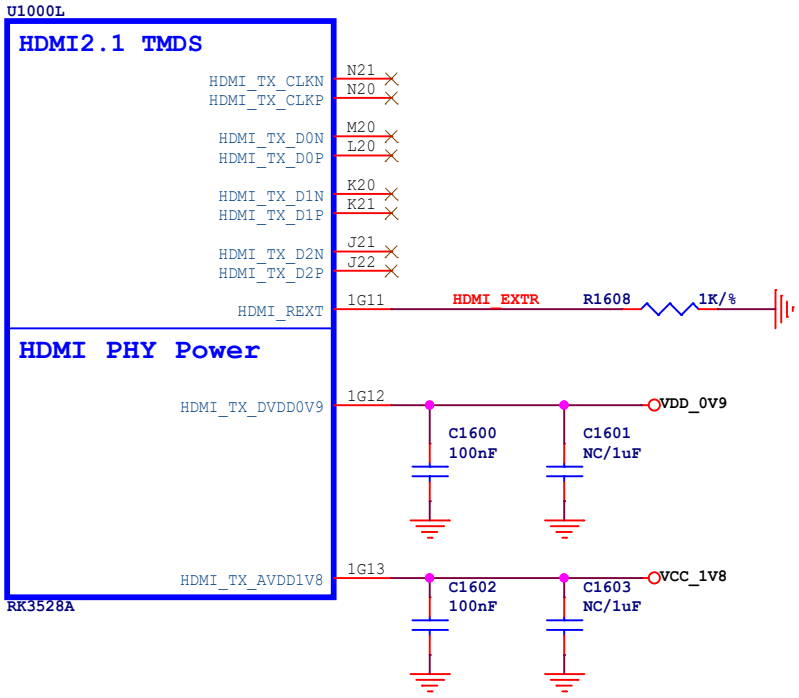
Item	Rup	Rdown	ADC	BOOT MODE
LEVEL1	DNP	100K	0	USB (Maskrom mode)
LEVEL2	100K	12K	114	
LEVEL3	100K	27K	228	FSPI--USB
LEVEL4	100K	51K	342	
LEVEL5	100K	82K	456	
LEVEL6	100K	120K	570	EMMC--USB
LEVEL7	100K	200K	683	EMMC--SD Card--USB
LEVEL8	100K	330K	796	SD Card--USB
LEVEL9	100K	820K	910	
LEVEL10	100K	DNP	1023	FSPI--EMMC--SD Card--USB





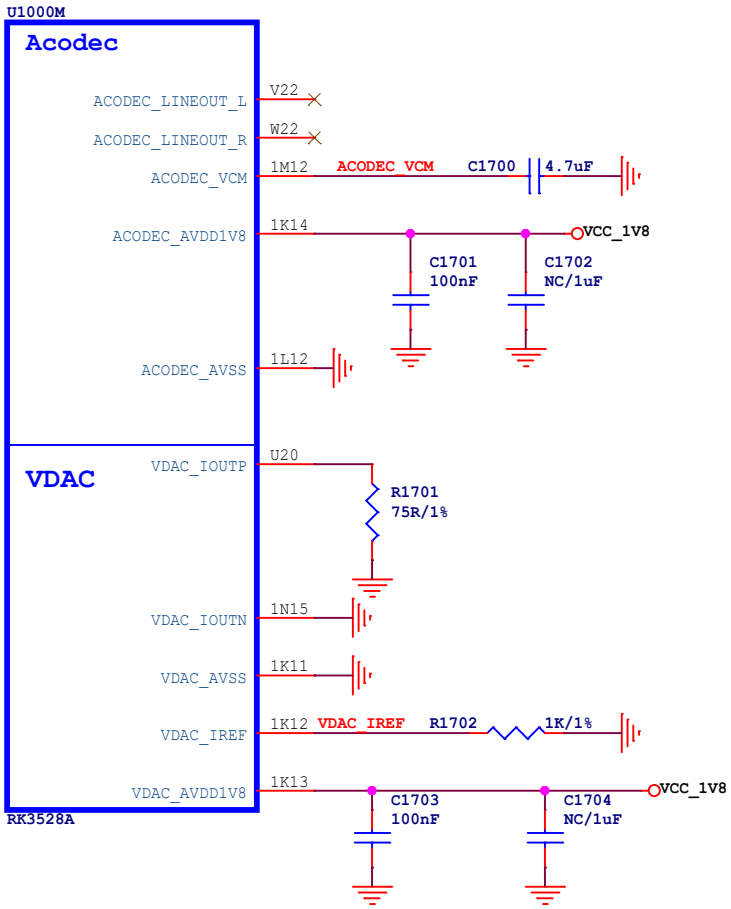
Size	Title: E20C		REV
A4	Page Name: 09.RK3528A SARADC/OTP		V1.10
Date: Thursday, June 06, 2024	Sheet 09 of 22		

RK3528A_L(HDMI PHY)



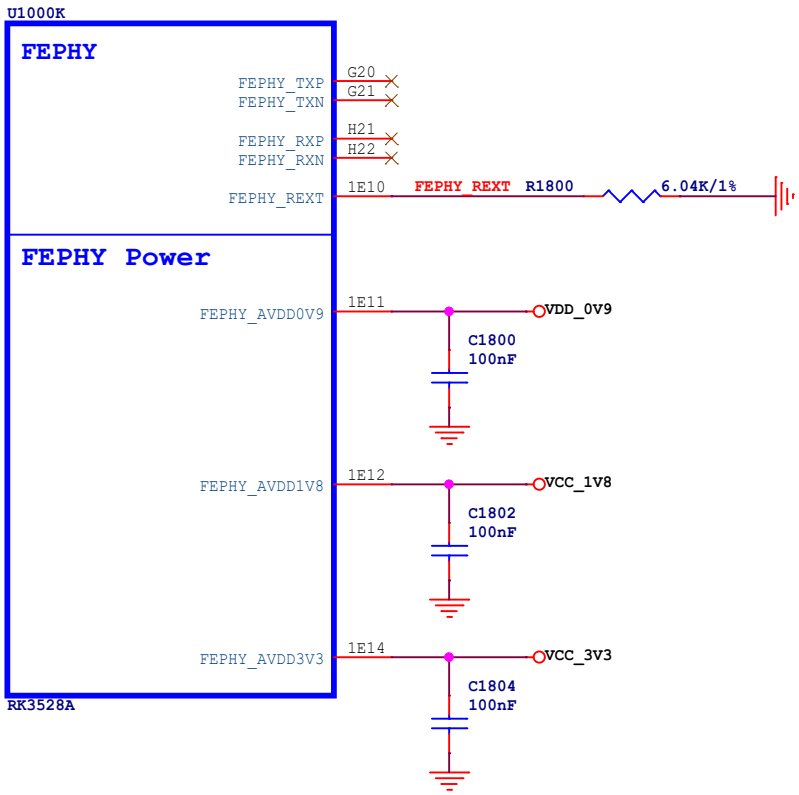
Size: A4	Title: E20C	REV: V1.10
	Page Name: 10.RK3528A HDMI Interface	
Date: Thursday, June 06, 2024	Sheet 10 of 22	

RK3528A_M(Acodec/VDAC)



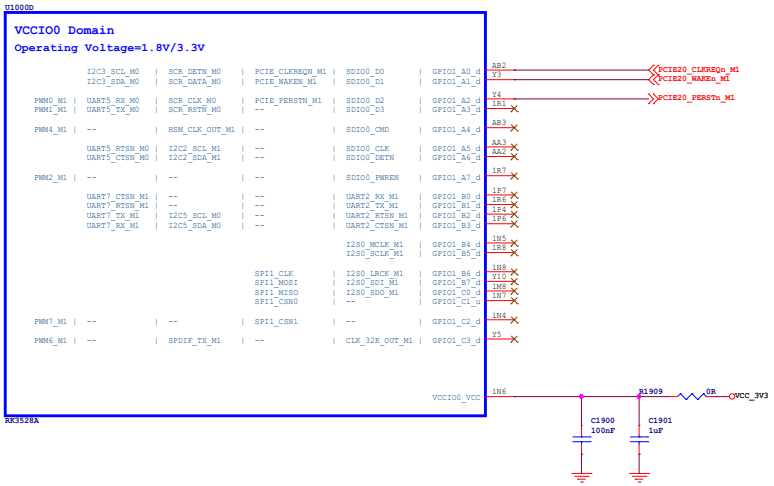
Size: A4	Title: E20C	REV: V1.10
	Page Name: 11.RK3528A AV OUT	
Date: Thursday, June 06, 2024	Sheet 11 of 22	

RK3528A_K(Embed FEPHY)

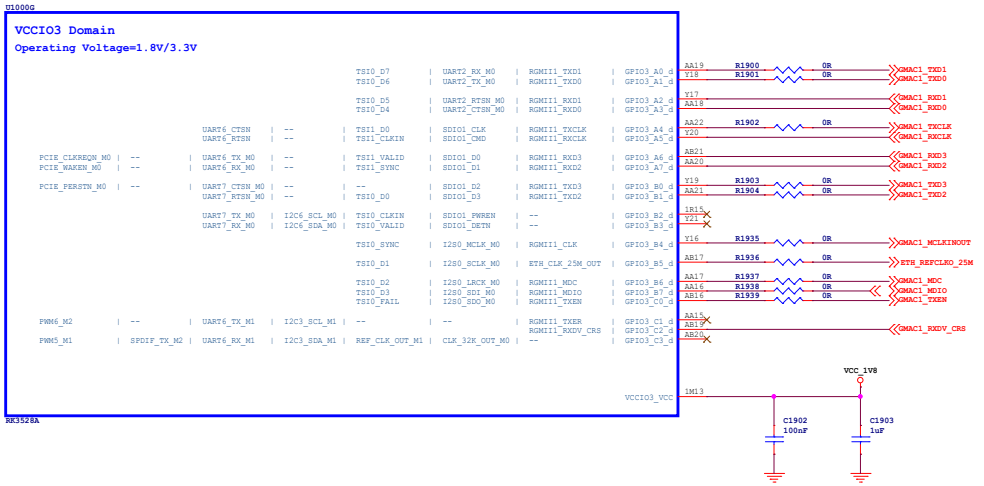


Size	Title:	E20C	REV
A4	Page Name:	12.RK3528A Embed FEPHY	V1.10
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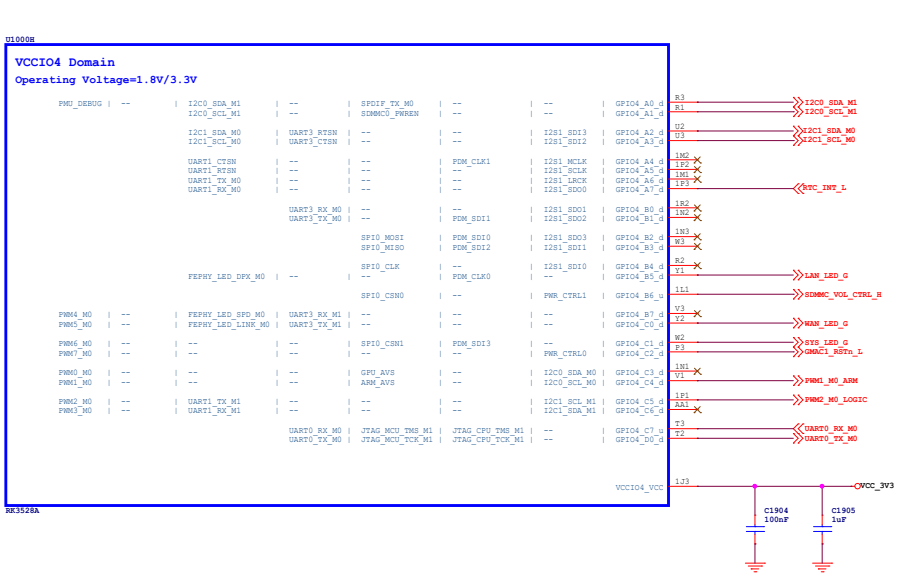
RK3528A_D(VCCIO0 Domain)

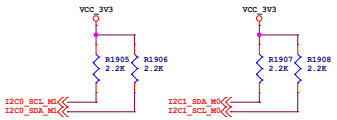


RK3528A_G(VCCIO3 Domain)

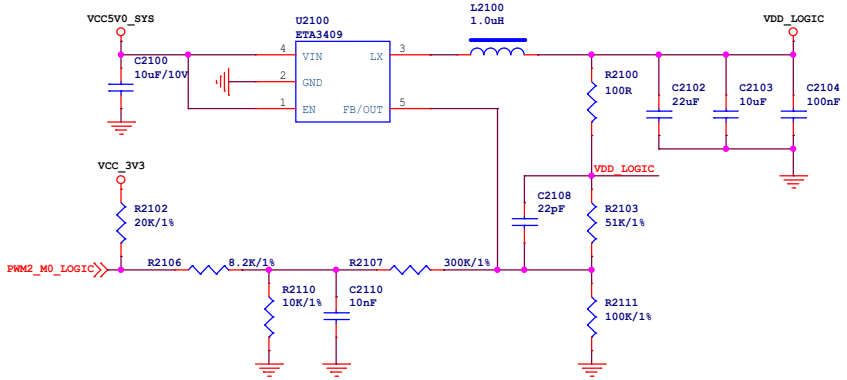


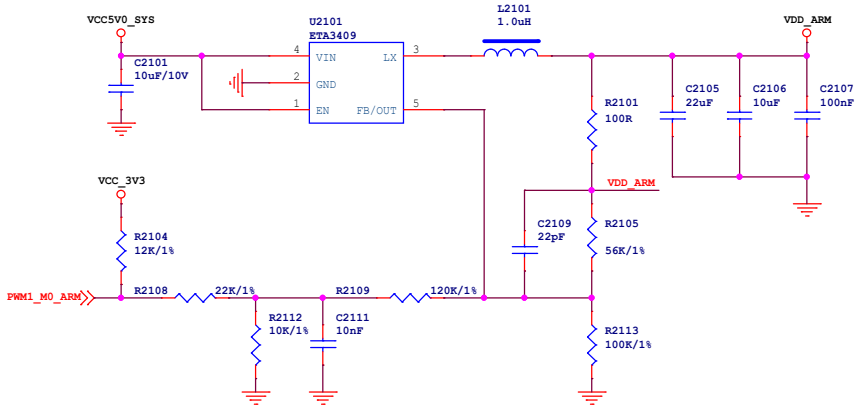
RK3528A_H(VCCIO4 Domain)

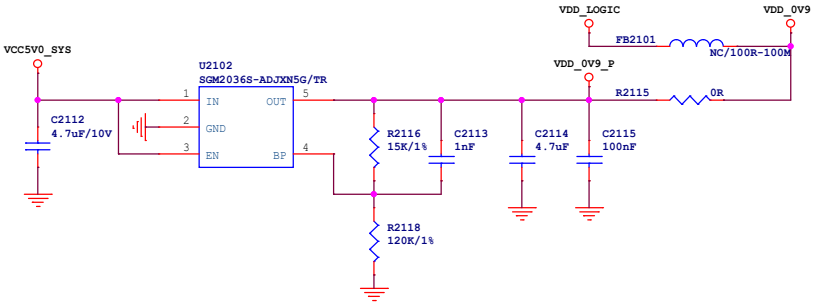


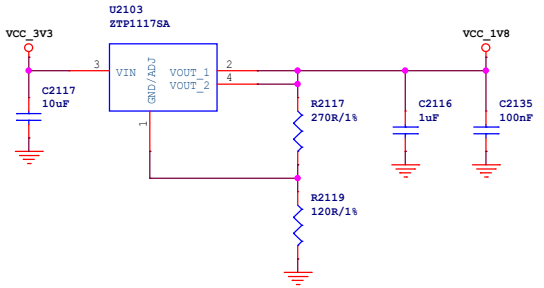


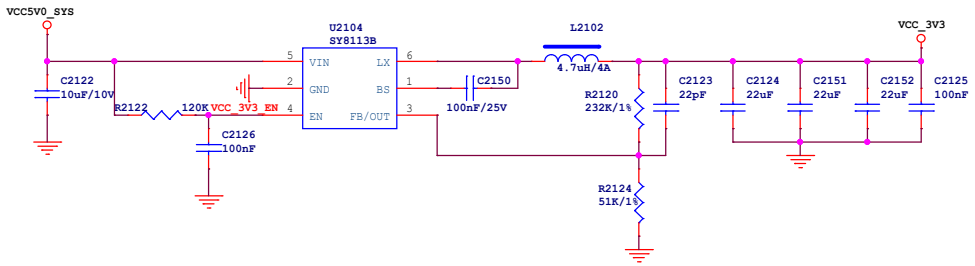
Size	Title: E20C		REV
A2	Page Name: 13.RK3528A_1.8V/3.3V GPIO		V1.10
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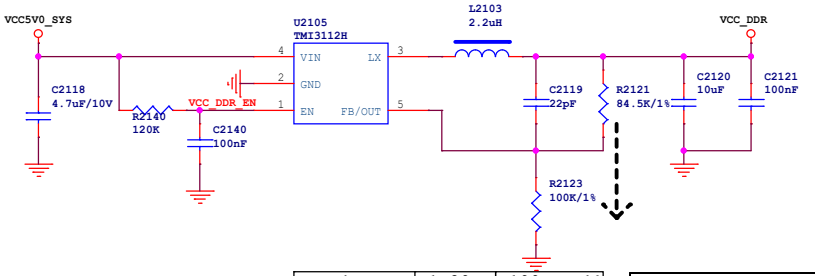










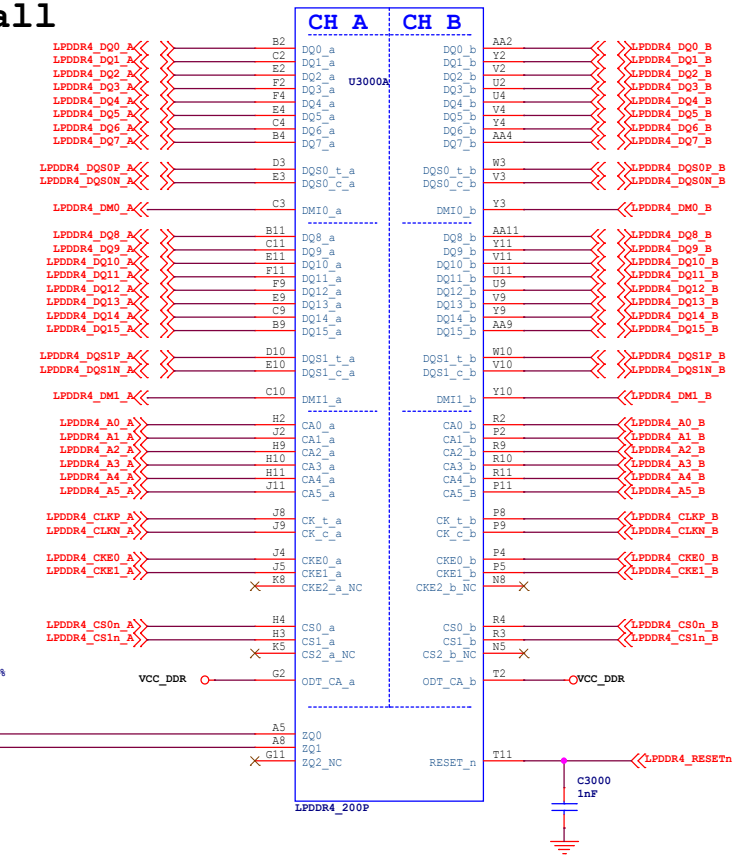


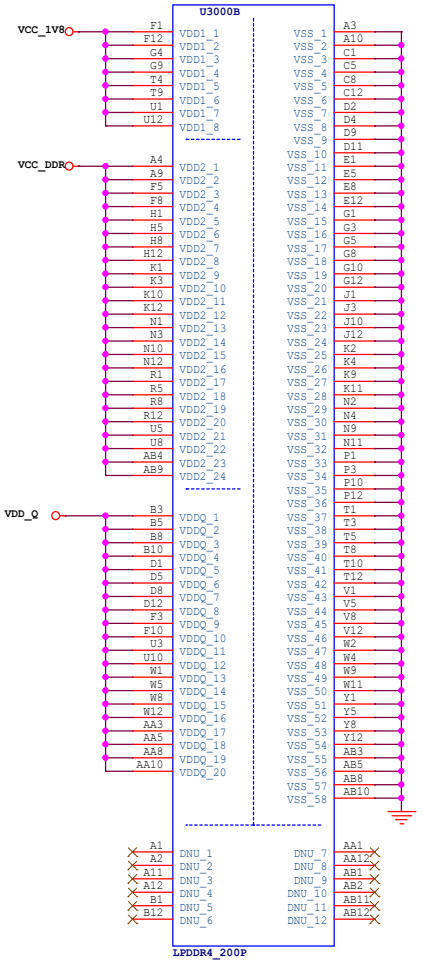
DDR4	1.20V	100K	1%
LPDDR4/4x	1.10V	84.5K	1%
DDR3	1.5V	150K	1%
DDR3L	1.35V	127K	1%
LPDDR3	1.25V	110K	1%

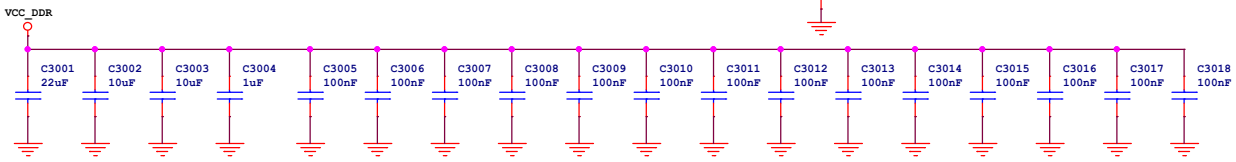
radxa

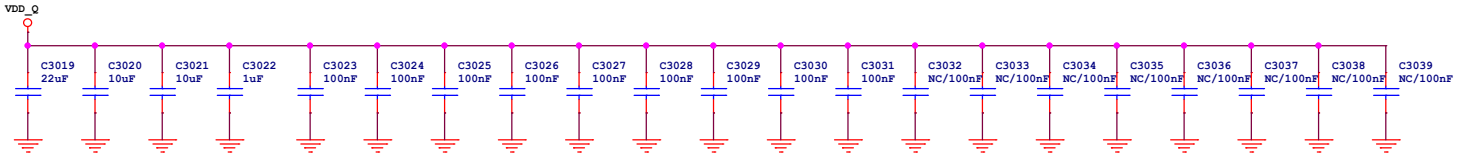
Size	Title: E20C	REV
A3	Page Name: 14.Power-DiscretePower	V1.10
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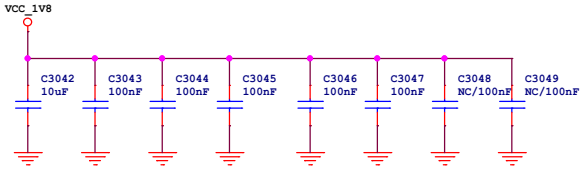
LPDDR4 200Ball



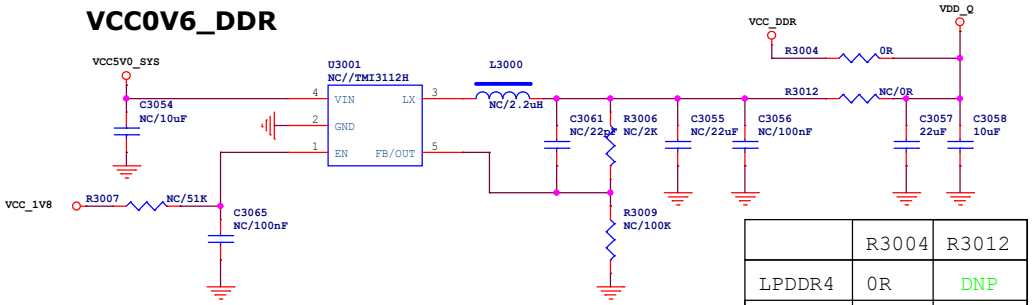








VCC0V6_DDR

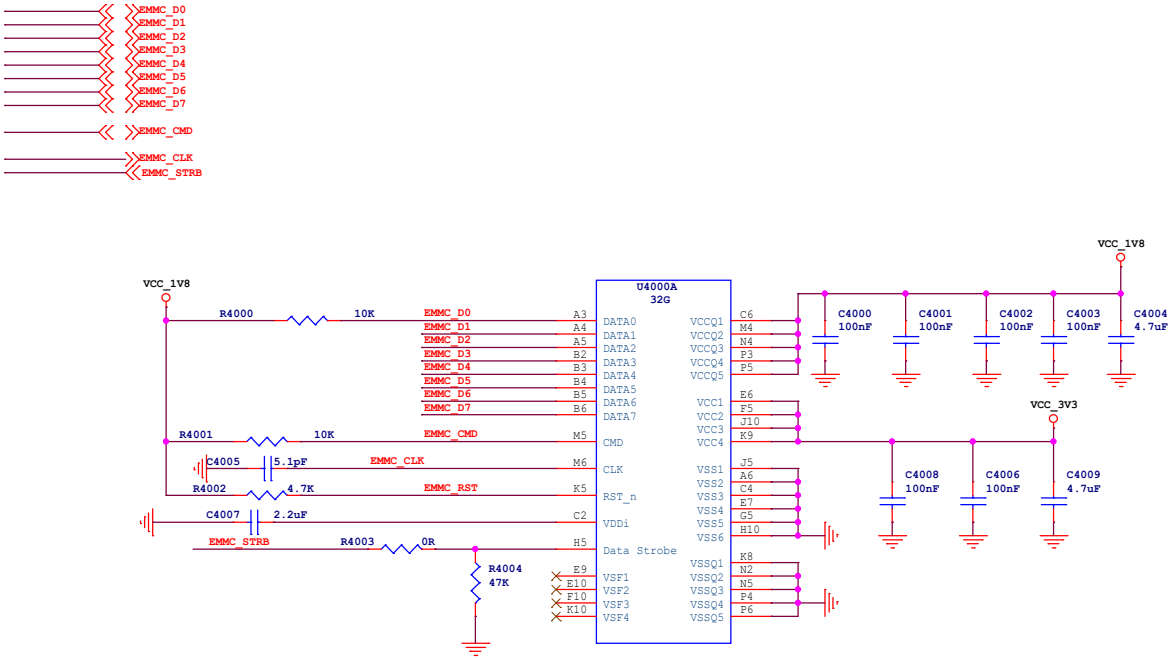


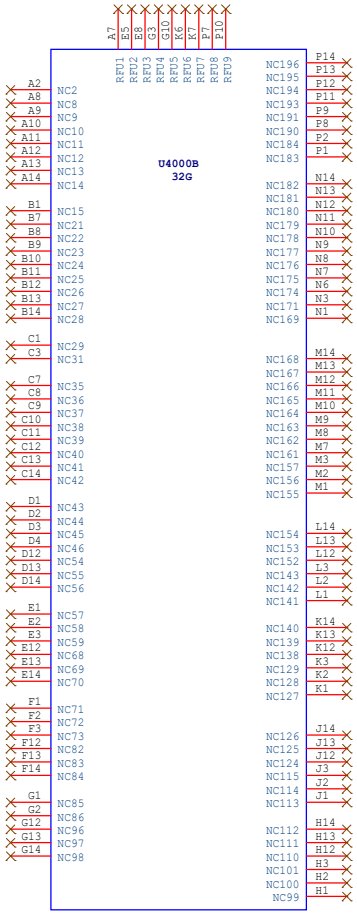
	R3004	R3012
LPDDR4	0R	DNP
LPDDR4X	DNP	0R

radxa

Size	Title:	REV
A3	E20C	V1.10
	Page Name: 15.DRAM-LPDDR4_1X32bit_200P	
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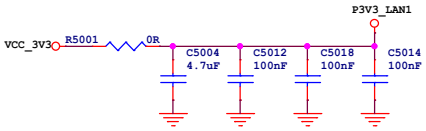
EMMC

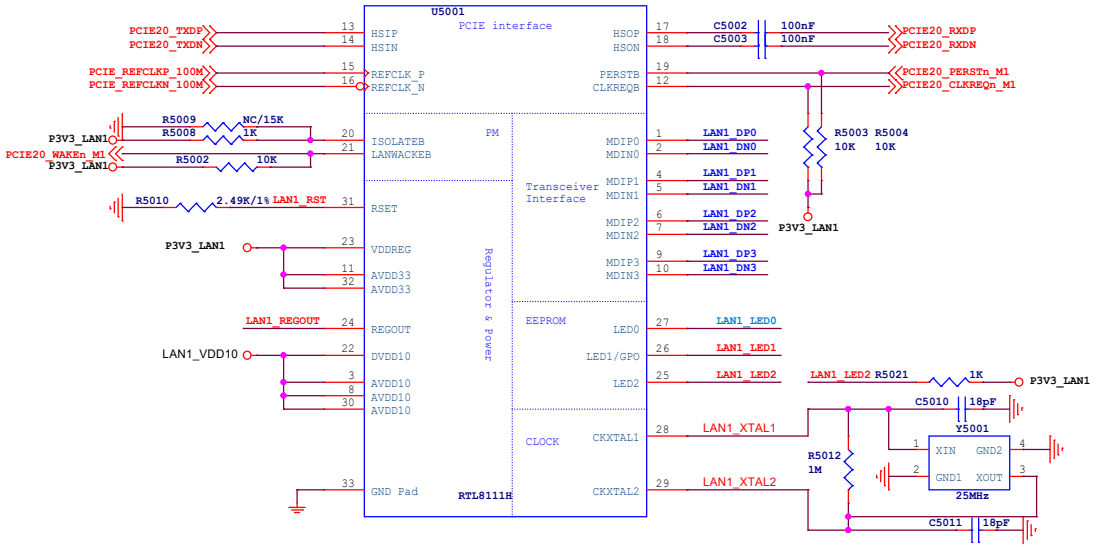


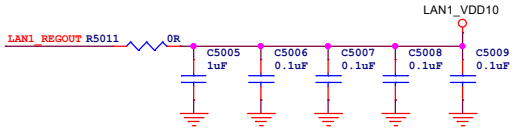


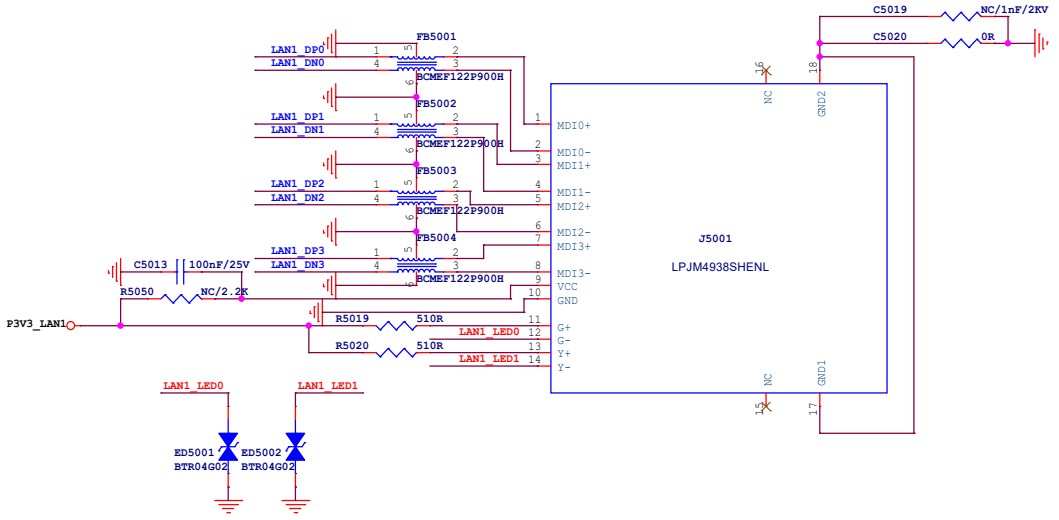
Size: A3	Title: E20C	REV: V1.10
	Page Name: 16.Flash-eMMC	
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LAN_RTL8111H

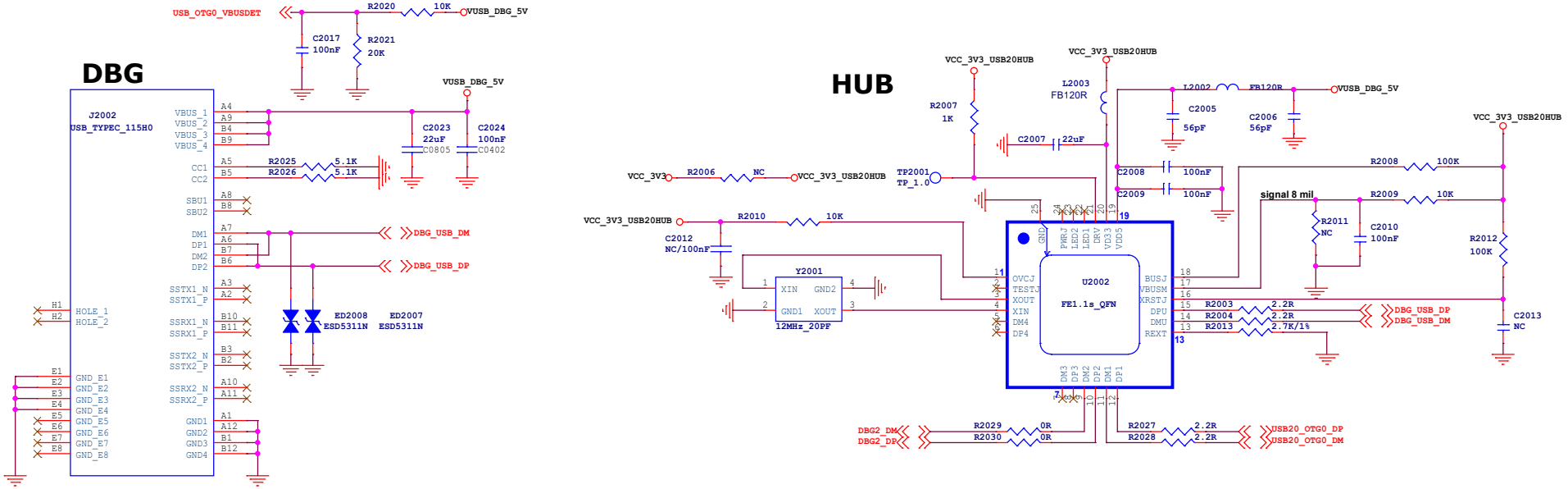




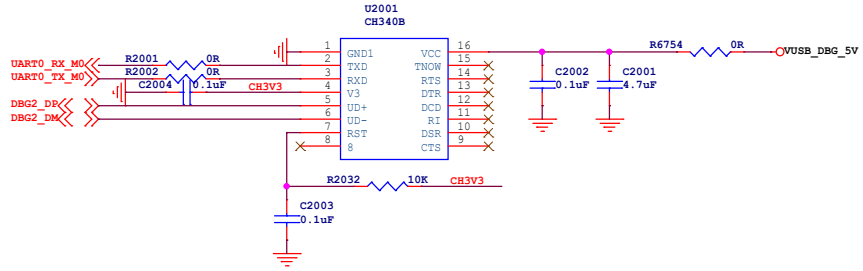




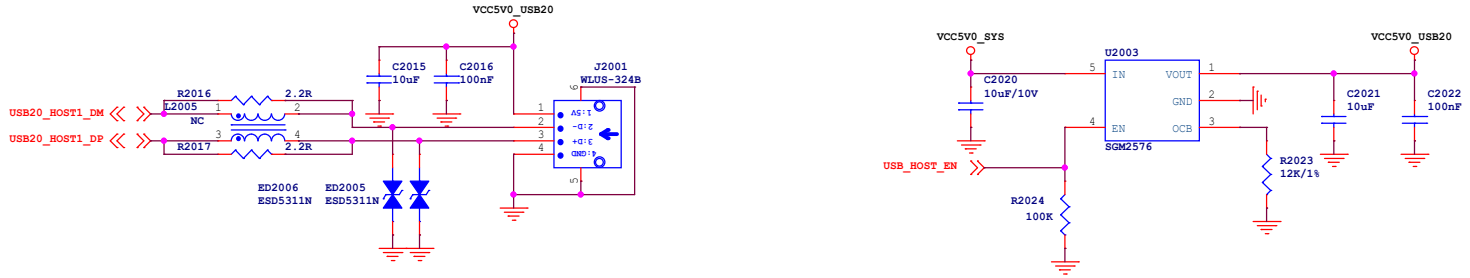
Size	Title: E20C		REV
A3	Page Name: 17.LAN_RTL8111H		V1.10
Date: Thursday, June 06, 2024	Sheet 17 of 22		



UART to USB



USB2.0



Size	Title: E20C		REV
A3	Page Name: 18.USB20_HUB		V1.10
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RGMII

GMAC1_TXD0
GMAC1_TXD1
GMAC1_TXD2
GMAC1_TXD3
GMAC1_TXEN
GMAC1_TXCLK
GMAC1_RXD0
GMAC1_RXD1
GMAC1_RXD2
GMAC1_RXD3
GMAC1_RXDV_CRS
GMAC1_RXCLK
ETH_REFCLKO_25M
GMAC1_MCLKINOUT
GMAC1_MDC
GMAC1_MDIO
GMAC1_RSTn_L

Y6700
PHY_XTALOUT
C6700 18pF
XIN GND2 GND1 XOUT
25MHz
C6701 18pF
PHY_XTALIN

ETH_REFCLKO_25M R6708 NC/0R PHY_XTALOUT

GMAC1_MCLKINOUT C6703 DNP R6710 100R PHY_CLKOUT125
R6712 120R

RJ45

PHY_MDI0+ PHY_MDI0-
PHY_MDI1+ PHY_MDI1-
PHY_MDI2+ PHY_MDI2-
PHY_MDI3+ PHY_MDI3-
FB6701 FB6702 FB6703 FB6704
BCMEF122P900H
C6726 100nF/25V
R6750 NC
VCC3V3_PHY
PHY_LED2/CFG_LDO1
PHY_LED1/CFG_LDO0
R6748 510R
R6713 510R
R6714 NC/0R
R6715 0R
R6716 NC/510R
J6701
LPJM4938SHENL
MDI0+ MDI0- MDI1+ MDI1- MDI2+ MDI2- MDI3+ MDI3- VCC GND G+ G- Y+ Y-
C6731 NC/1nF/2KV
C6730 0R
NC GND2 GND1

VDD10_PHY
R6722 2.49K
VCC3V3_PHY
T6700 TP_1.0
U6700
RTL8211F
E_Pad
AVDD33 RSTO AVDD10 XTAL_IN/EXT_CLKIN XTAL_OUT CLKOUT LED2/CFG_LDO1 LED1/CFG_LDO0 LED0/CFG_EXT INTB/PMEB
MDI[0]+ MDI[0]- AVDD10 MDI[1]+ MDI[1]- MDI[2]+ MDI[2]- AVDD10 MDI[3]+ MDI[3]-
REG_OUT DVDD33 DVDD_RG RXC/PHYAD1 RXCTL/PHYAD2 RXD0/RXDLY RXD1/TXDLY RXD2/PLLOFF RXD3/PHYAD0 DVDD10
AVDD33 PHYRSTB MDC MDIO TXD3 TXD2 TXD1 TXD0 TXCTL TXC
PHY_RSTn GMAC1_MDC GMAC1_MDIO GMAC1_TXD3 GMAC1_TXD2 GMAC1_TXD1 GMAC1_TXD0 GMAC1_TXEN GMAC1_TXCLK
C6710 NC/3pF
R6736 NC/4.7K
R6737 1.8K
C6717 100nF
GMAC1_RSTn_L R6738 0R

PHY_RXD0/RXDLY R6739 22R GMAC1_RXD0
PHY_RXD1/TXDLY R6740 22R GMAC1_RXD1
PHY_RXD2/PLLOFF R6741 22R GMAC1_RXD2
PHY_RXD3/PHYAD0 R6742 22R GMAC1_RXD3
PHY_RXCLK/PHYAD1 R6743 22R GMAC1_RXCLK
PHY_RXDV/PHYAD2 R6744 22R GMAC1_RXDV_CRS
C6723 DNP

R6717 4.7K PHY_LED0/CFG_EXT R6718 NC/4.7K VCC3V3_PHY
R6719 NC/4.7K PHY_LED2/CFG_LDO1 R6720 4.7K
R6721 4.7K PHY_LED1/CFG_LDO0

R6723 NC/4.7K PHY_RXD3/PHYAD0 R6724 4.7K VCCIO_PHY
R6725 4.7K PHY_RXCLK/PHYAD1
R6726 4.7K PHY_RXDV/PHYAD2

PHY Address	PHYAD[2:0]
1 (default)	3'b001

R6727 NC/4.7K PHY_RXD0/RXDLY R6728 4.7K VCCIO_PHY
R6729 4.7K PHY_RXD1/TXDLY R6730 NC/4.7K VCCIO_PHY
R6732 NC/4.7K PHY_RXD2/PLLOFF R6733 4.7K VCCIO_PHY

PHY_REG_OUT L6700 2.2uH
VDD10_PHY
C6714 4.7uF C6715 100nF C6716 100nF
C6720 100nF C6724 100nF C6725 100nF

PHY_LED2/CFG_LDO1 C6705 NC/100pF
PHY_LED1/CFG_LDO0 C6706 NC/100pF

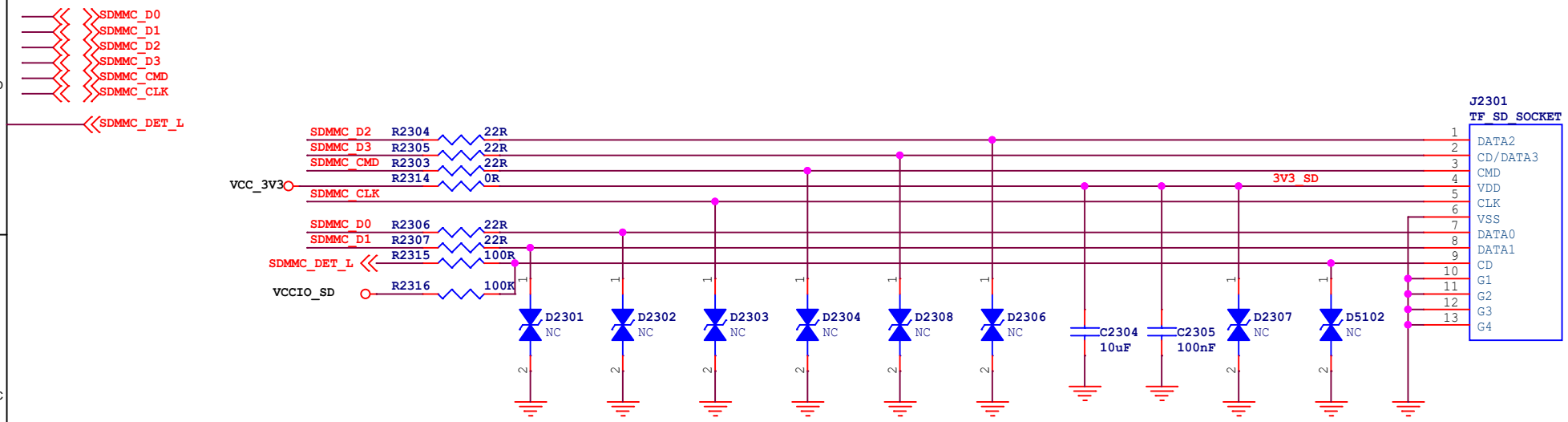
VCC_1V8 R6731 NC/0.1R
VCC_3V3 R6734 NC/0.1R
VCCIO_PHY
C6707 4.7uF C6708 1uF C6709 100nF

VCC_3V3 R6735 0.1R
VCC3V3_PHY
C6711 4.7uF C6712 100nF C6713 100nF
C6718 4.7uF C6719 100nF

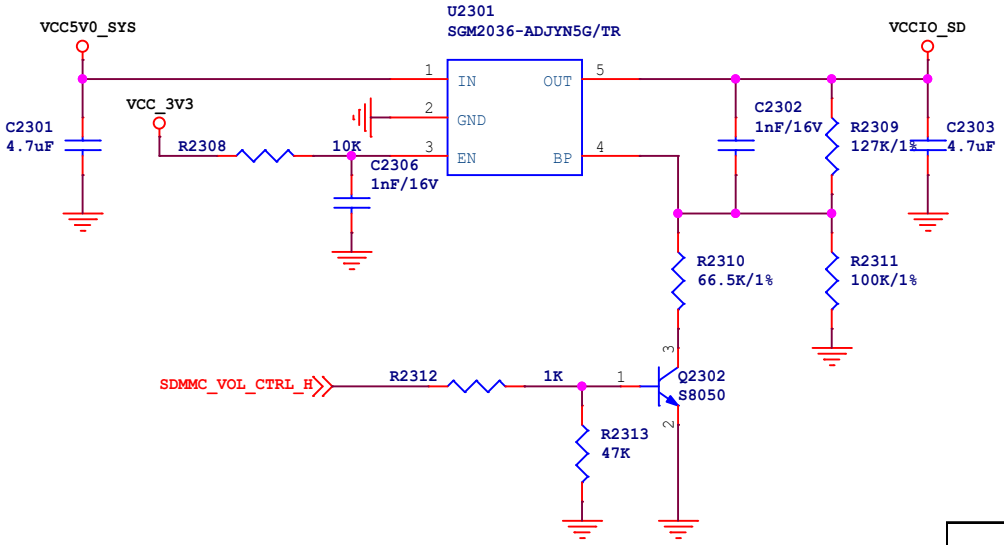
radxa

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	Page Name: 19.Ethernet_RGMII	
Date: Thursday, June 06, 2024	Sheet 19 of 22	

Micro_SD Card

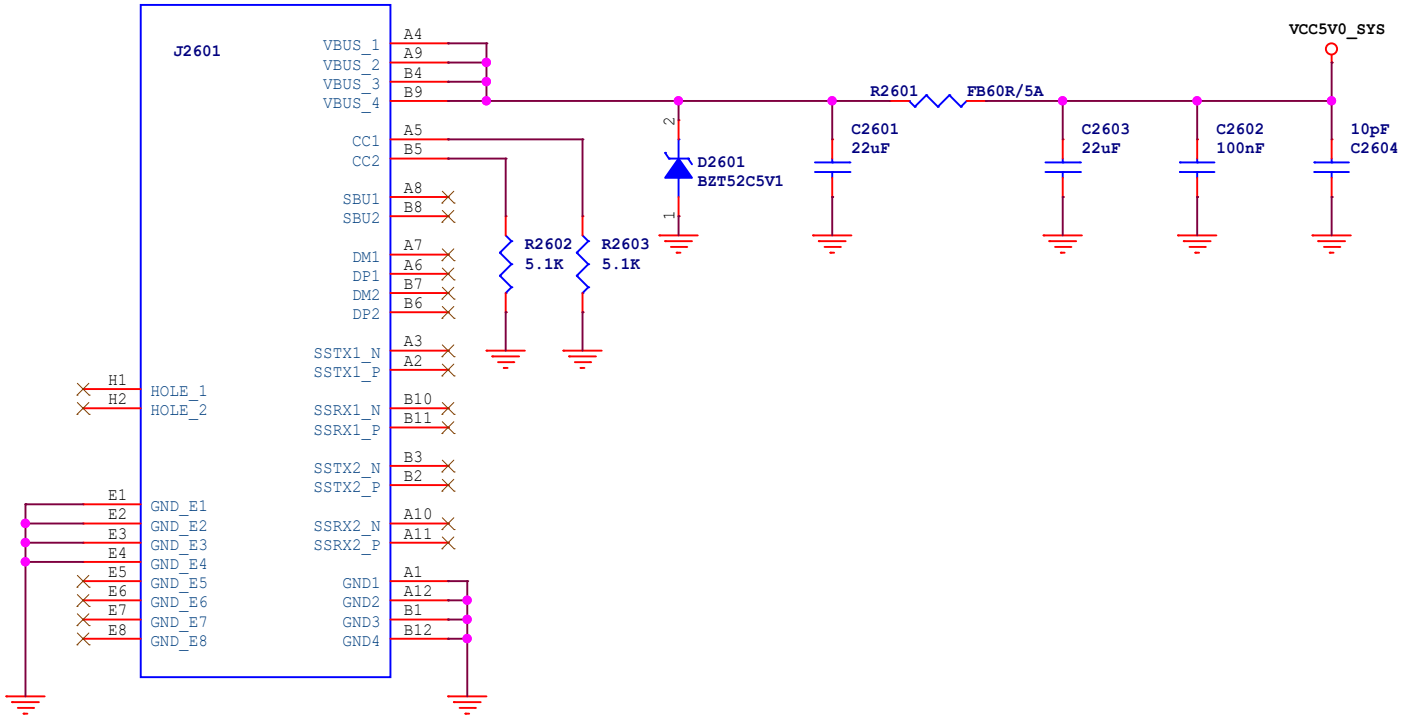


VCCIO_SD



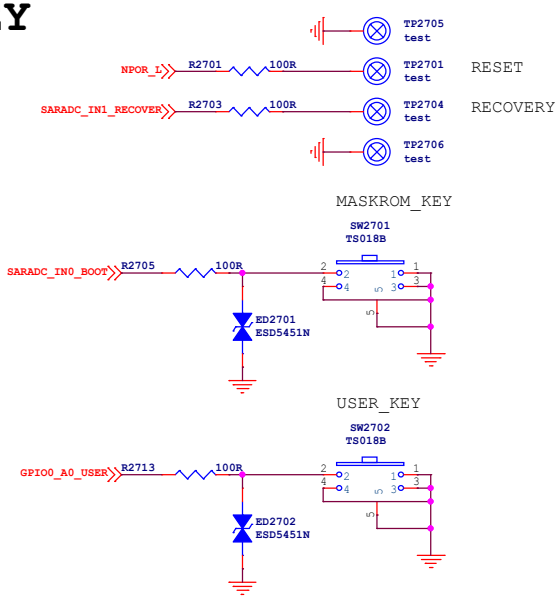
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	Page Name: 20.Micro_SD		
Date: Thursday, June 06, 2024		Sheet 20 of 22	

DC 5V

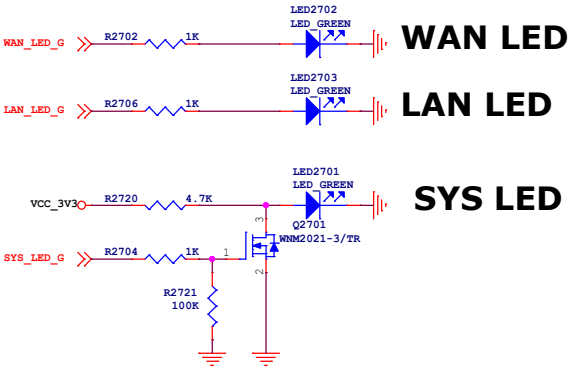


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	Page Name: 21.5V_Power	
Date: Thursday, June 06, 2024	Sheet 21 of 22	

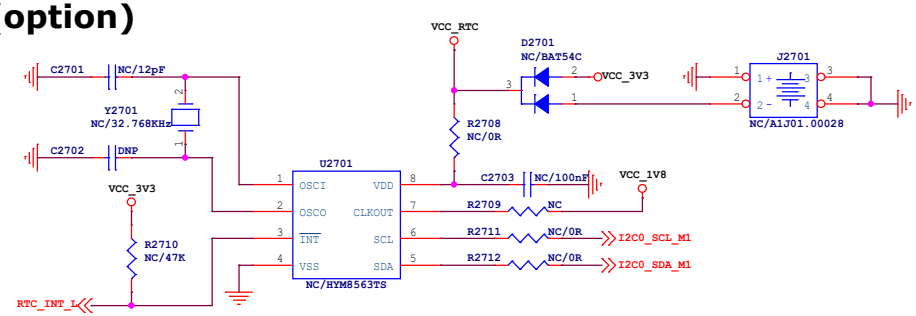
KEY



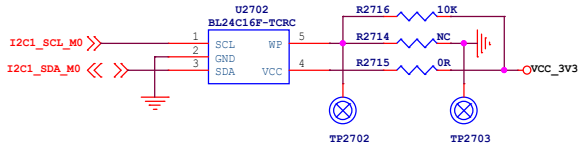
LED



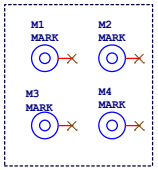
RTC(option)



EEPROM



MARK



Size: A3	Title: E20C	REV: V1.10
	Page Name: 22.Key_LED	
Date: Thursday, June 06, 2024	Sheet 22 of 22	